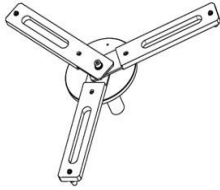
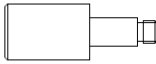
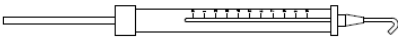
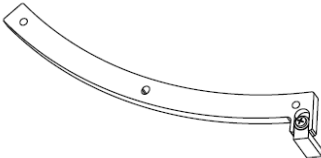


Reaction gear adjustment

Reaction gear adjustment	1
1. Adjustment tool.....	2
1.1 Making the checking roll.....	2
2. Preparation.....	3
3. Reaction gear adjustment.....	6
3.1 Measurement of Reaction gear rotation torque	7
3.2 Adjustment of Moter unit position.....	10
3.3 Adjustment of Reaction gear position	13
3.4 Adjustment(check) of count detection sensor position	16
4. Accomplishment	18

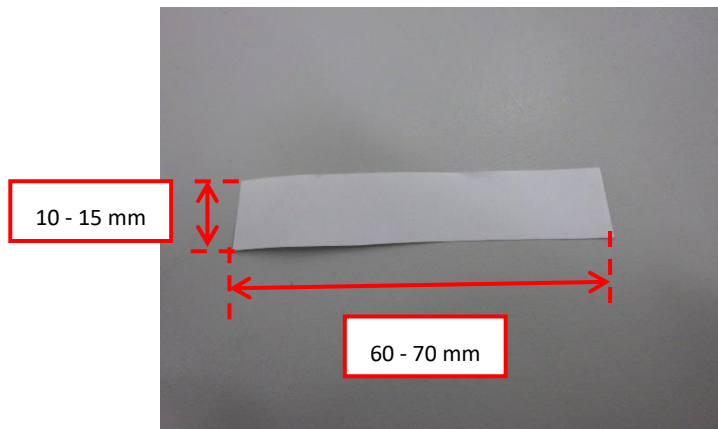
1. Adjustment tool

The following adjustment tools are used for this adjustment. Please refer to "S03-01 List of adjustment tools" for a list of jigs.

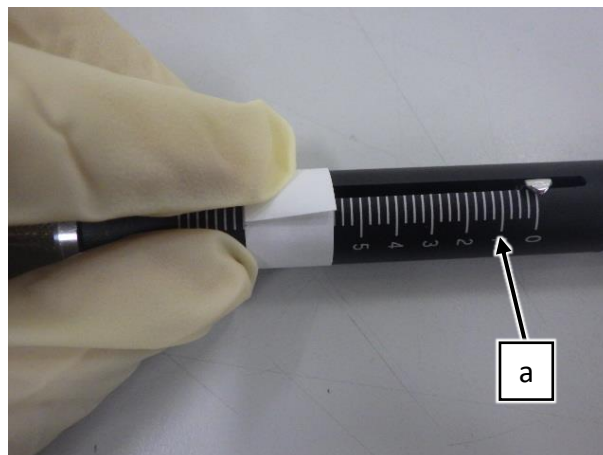
P/N	Name	Appearance
848-3870	TOOL 703 GEAR ADJUSTMENT	
734-1465	GEAR PIN	
734-3996	SPRING BALANCE	
848-3780	DETECTOR ADJUSTMENT T	

1.1 Making the checking roll

- (1) Prepare a paper (width: 10 to 15 mm, length: 60 to 70 mm)

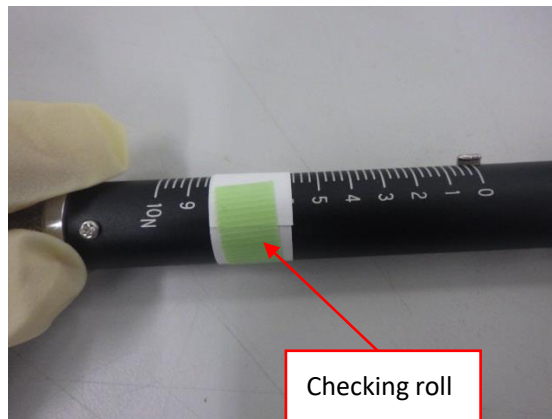


- (2) Roll the paper around the tool (a).



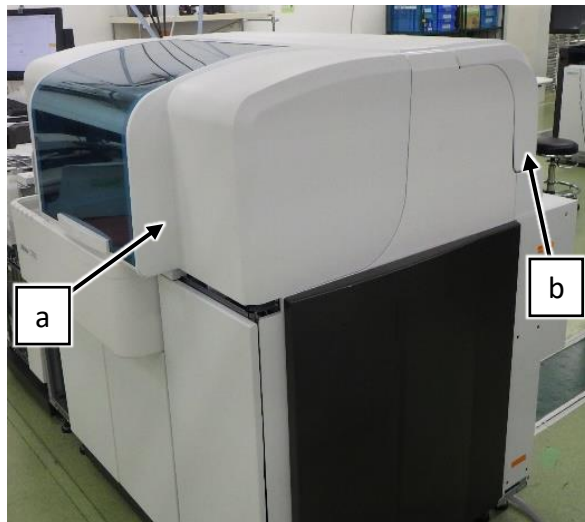
(a): SPRING BALANCE (P/N: 734-3996)

- (3) Fix the paper with a tape.

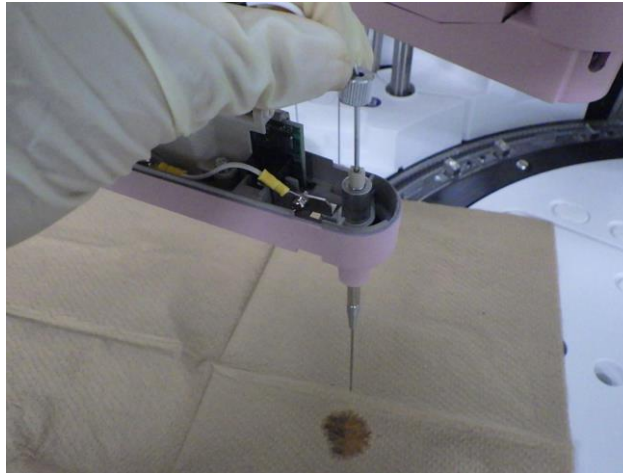


2. Preparation

- (1) Turn off c 703 (refer to "S2-13 AU and SB powered off procedure").
- (2) Open COVER TOP F ASSY (a) COVER TOP R ASSY (b).



- (3) Remove Reagent probe R1B (refer to “Replacing the reagent probe” in User guide).

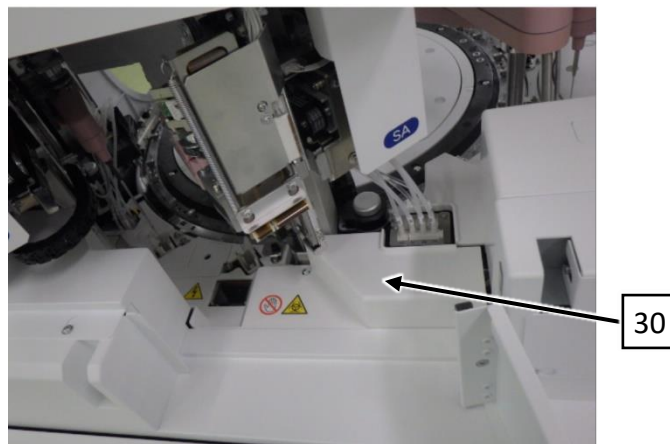
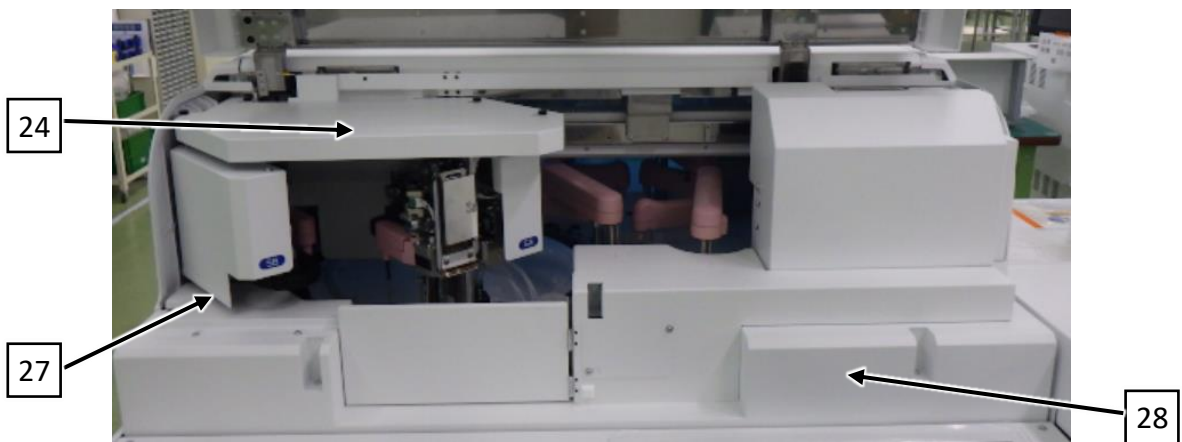
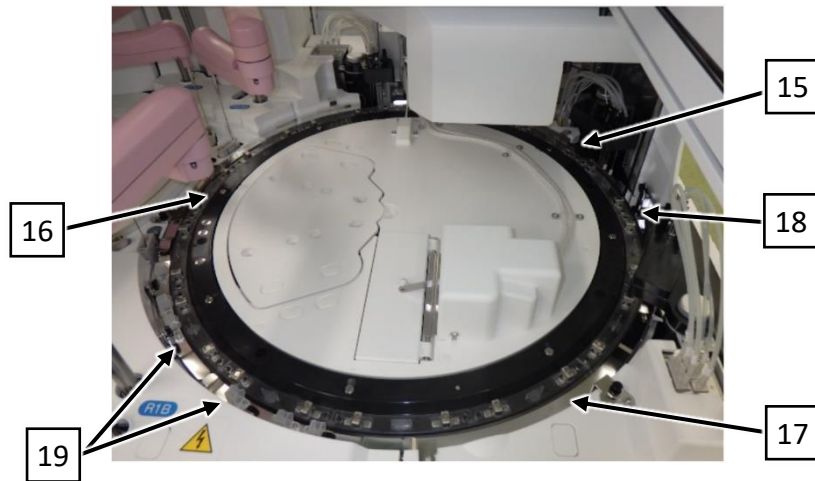


- (4) Remove Reagent probe R23B (refer to “Replacing the reagent probe” in User guide).

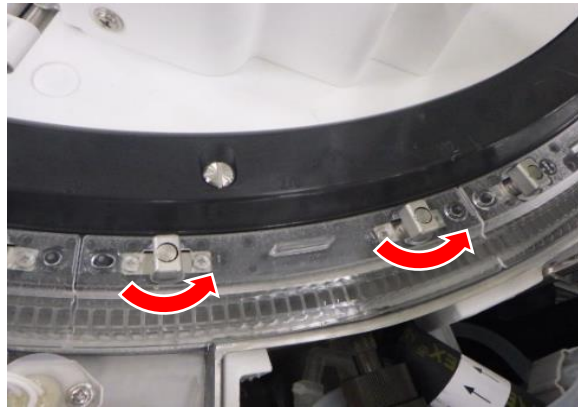


- (5) Remove No. 14, 15 16, 17, 18, 19 24, 27, 28, 30 covers (refer to “S06-41 Removal and attachment of covers”).





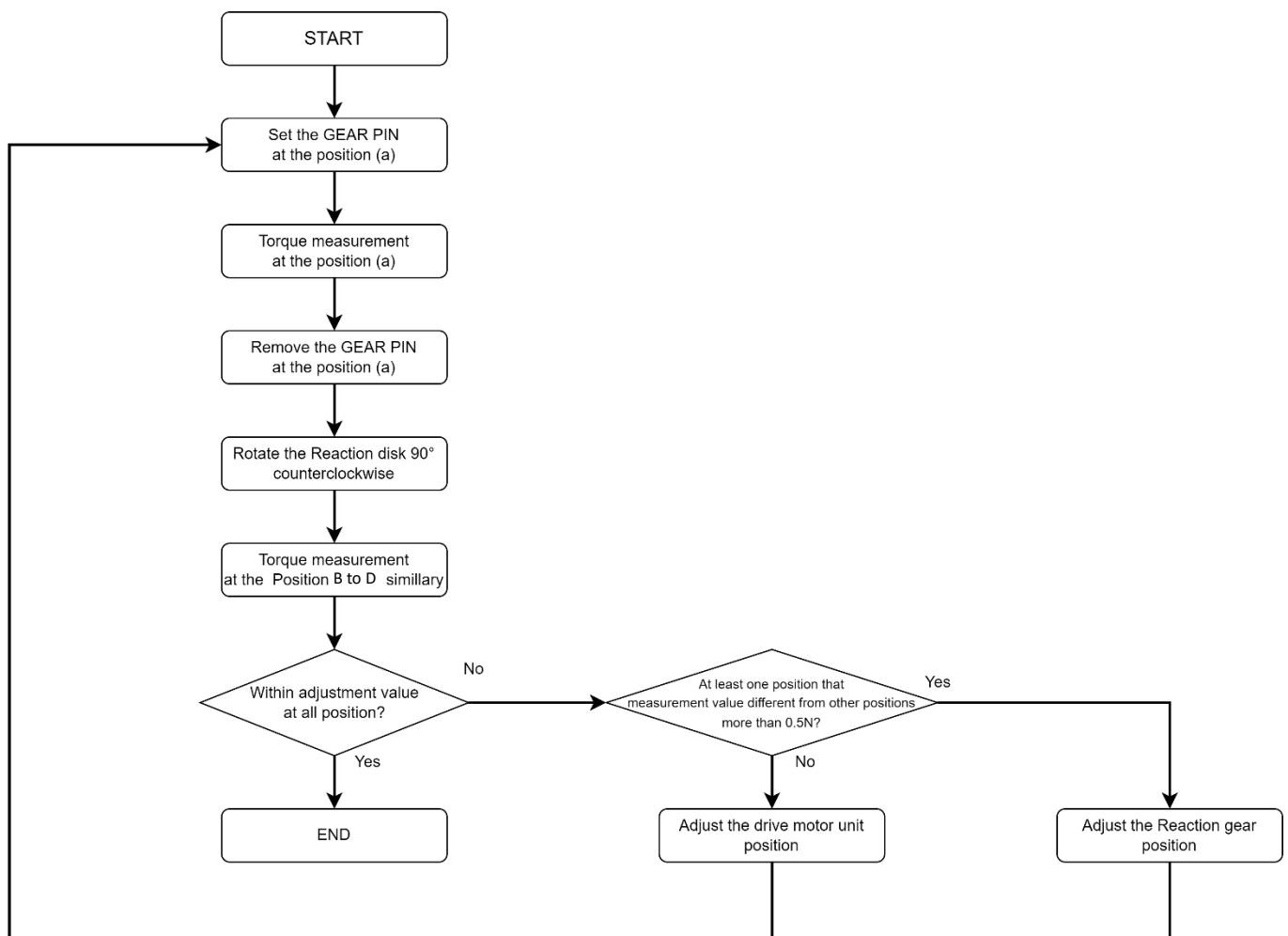
- (6) Twist the 2 quick screws and remove the cell. Similarly remove all the other cells.



- (7) Remove Reaction Disk (refer to “S06-07 Replacement procedure for Reaction disk”).

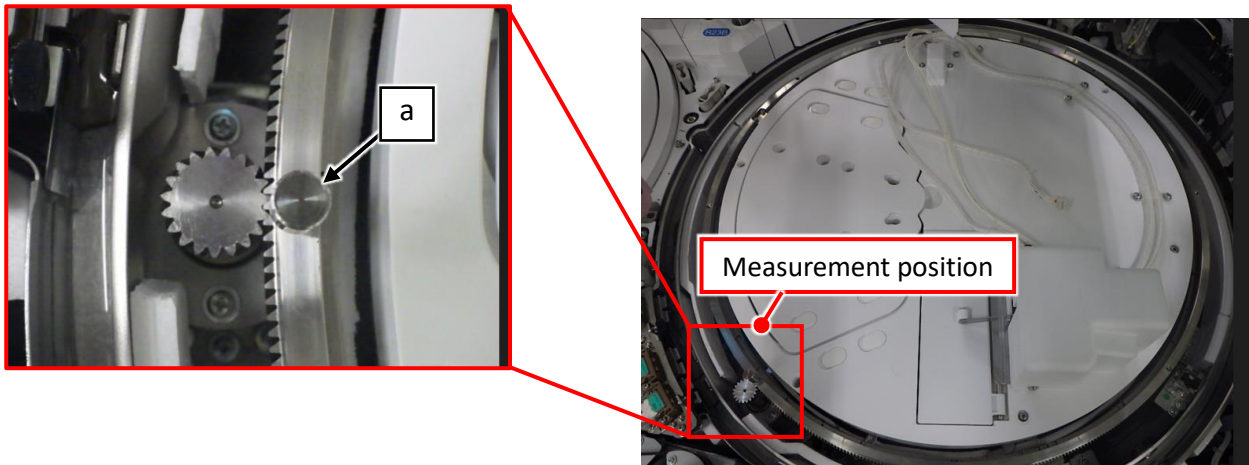
3. Reaction gear adjustment

The Reaction gear adjustment procedure is shown in the flowchart below.



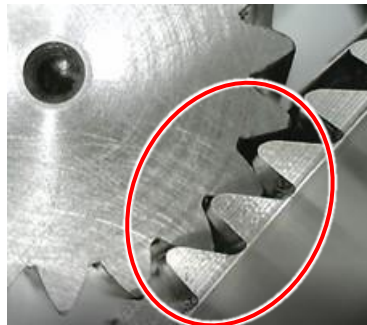
3.1 Measurement of Reaction gear rotation torque

- (1) Set the pin (a) at the measurement position on Reaction gear.



(a): GEAR PIN (P/N:734-1465)

- (2) Make sure that there is no gap between the teeth of the Reaction gear and the Drive motor gear unit when rotate the Reaction gear.



- (3) Hook the tip of the tool (a) to the upper part of the pin (b).



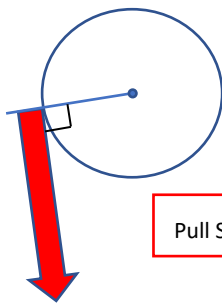
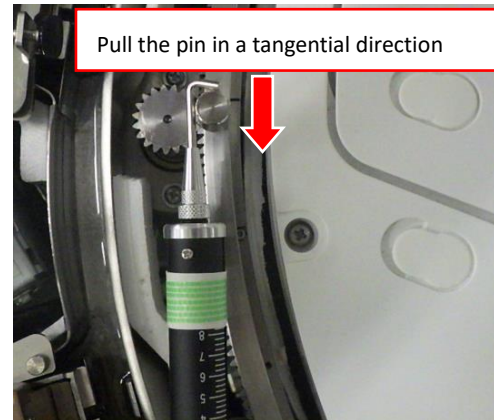
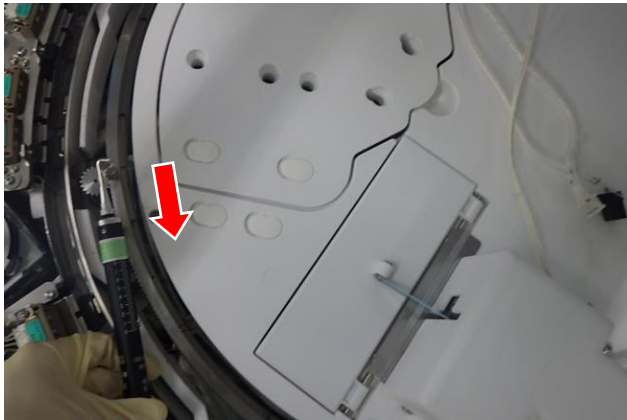
(a): SPRING BALANCE (P/N: 734-3996)

(b): GEAR PIN (P/N: 734-1465)

- (4) Pull the pin (a) slowly using the tool (b) in a tangential and horizontal direction and check the measurement value when Reaction gear just begins to move.

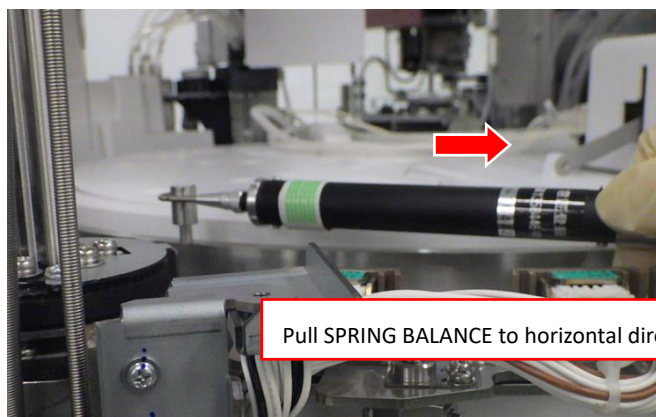
Note: The method to check the rotation torque is different from c 702. Check measurement values at the end of the checking roll (c).

【Top view of Reaction gear】



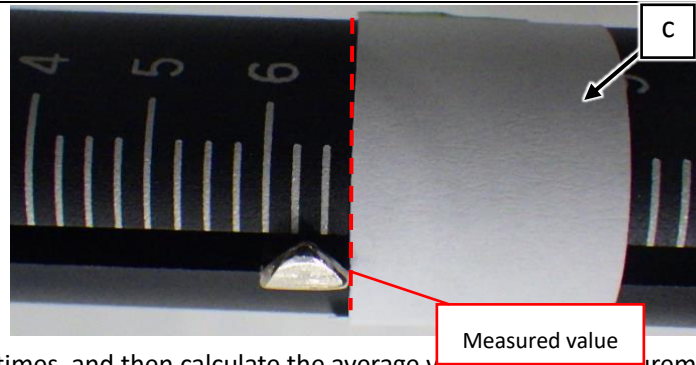
Pull SPRING BALANCE in a tangential direction

【Side view of Reaction gear】

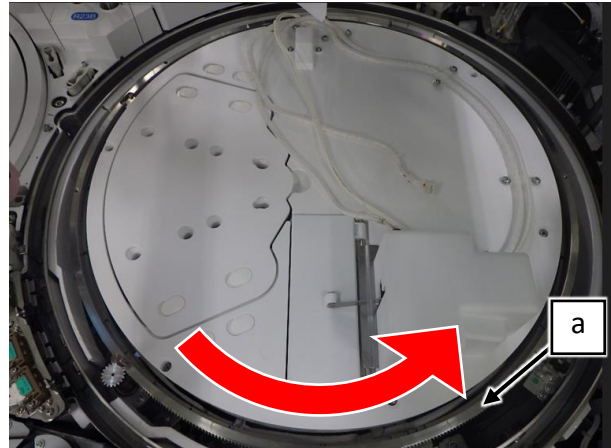


(a): GEAR PIN (P/N: 734-1465)

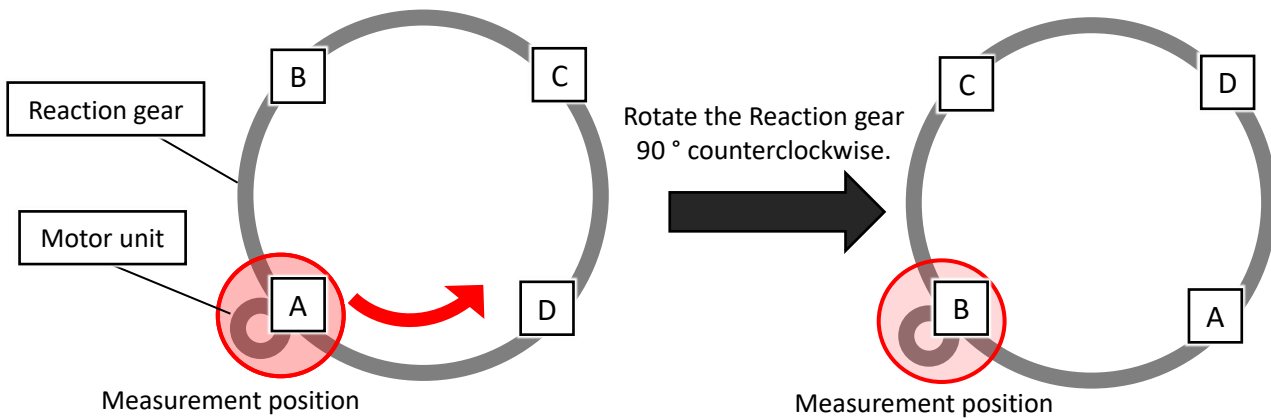
(b): SPRING BALANCE (P/N: 734-3996)



- (5) Repeat the step (4) 3 times, and then calculate the average value from 3 measurement values.
- (6) Rotate Reaction gear (a) 90 ° counterclockwise.

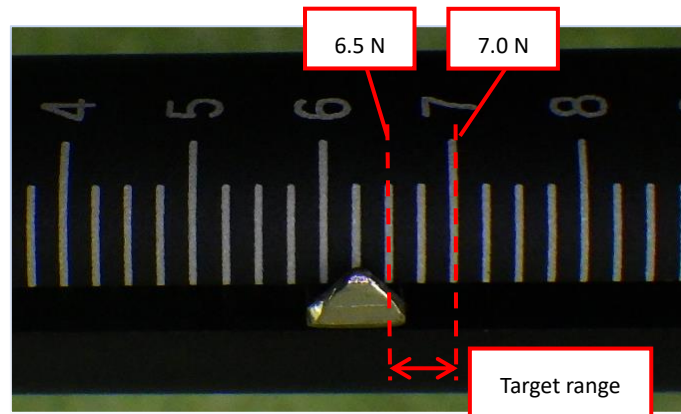


- (7) Relocate the pin at the next measurement position near Motor unit.



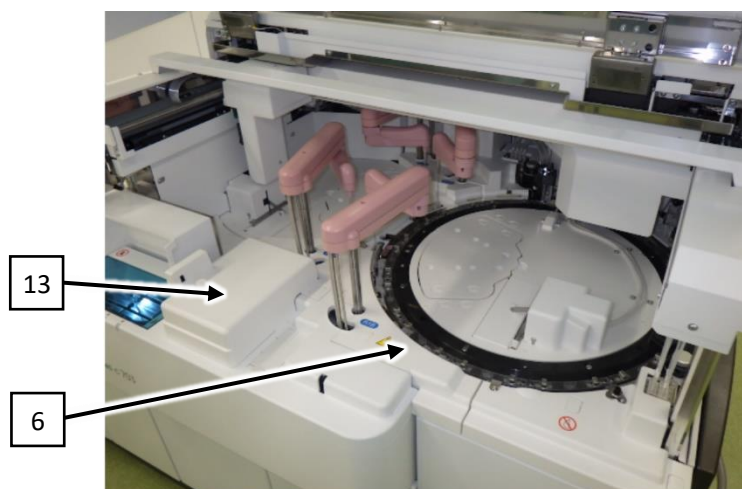
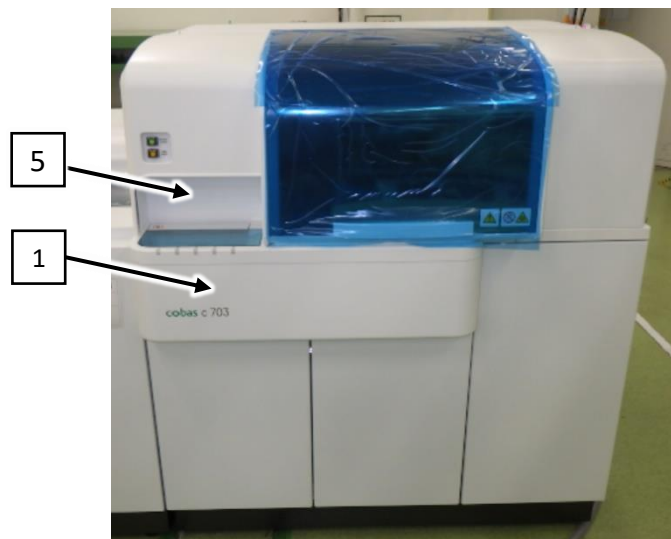
- (8) Perform the steps (2) to (7) at each of positions B to D.

- (9) Make sure that each average of measurement values at the 4 positions are within the target range of 6.5 N to 7.0 N. If, measurement values are within the target range of 6.5 N to 7.0 N, proceed to the chapter 4.
- If the difference between the maximum and minimum of the averages is more than 0.5 N, proceed to the section 3.3.
- If the difference is less than 0.5 N and all averages at the 4 positions are not within the target range of 6.5 N to 7.0 N, proceed to the section 3.2.

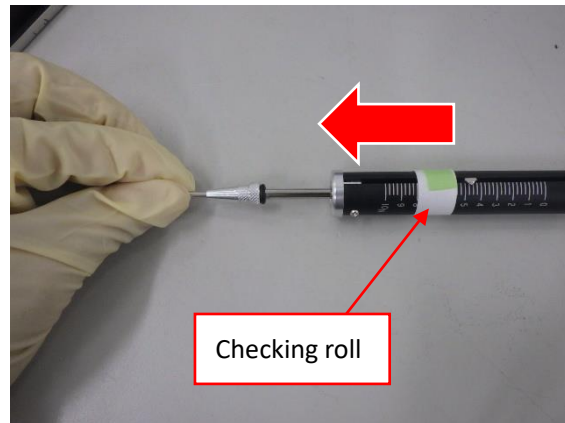
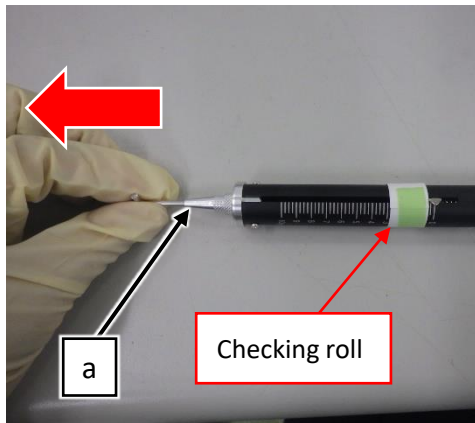


3.2 Adjustment of Moter unit position

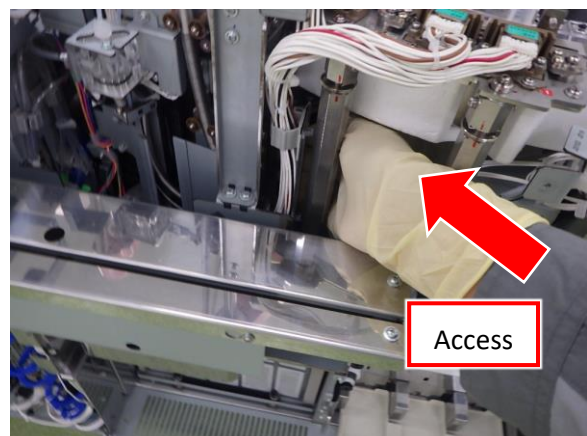
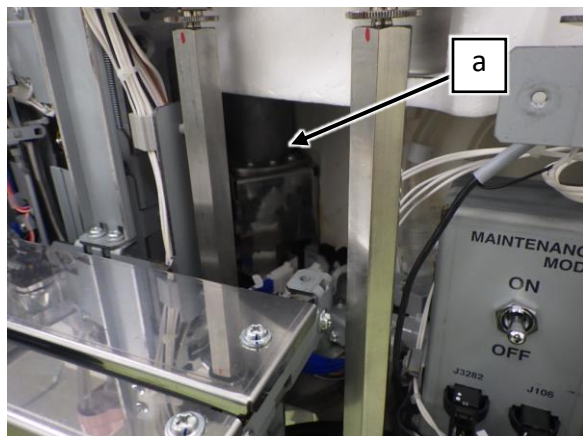
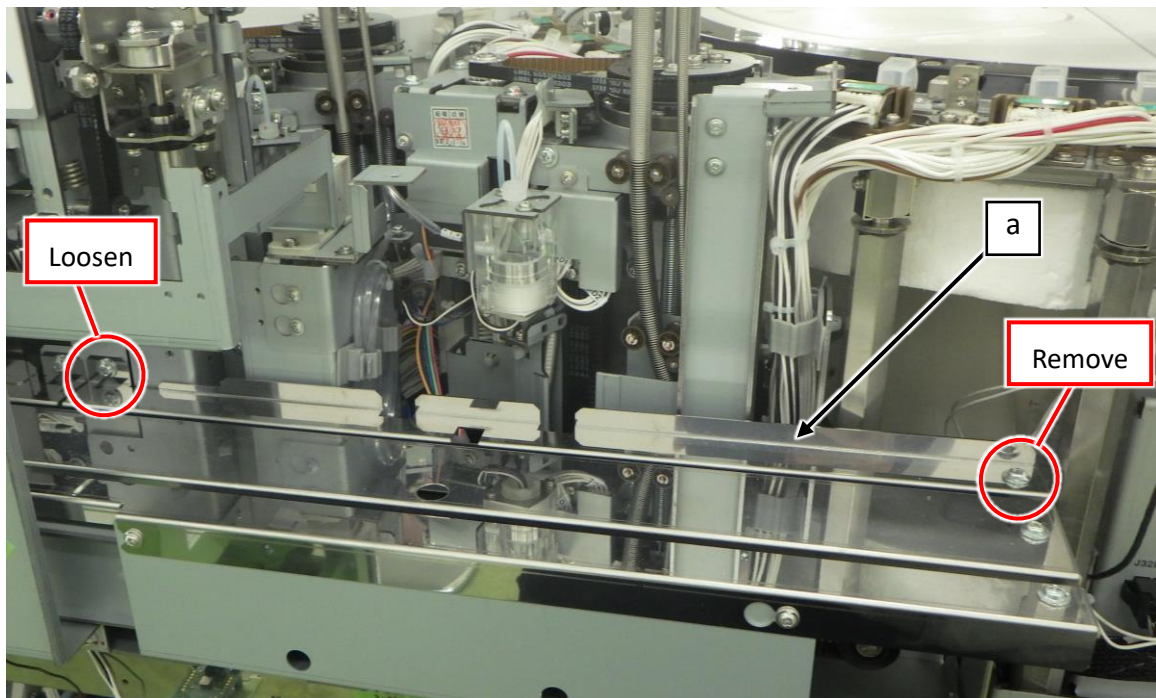
- (1) Remove No.1, No.5, No.6, No.13 covers (refer to "S06-41 Removal and attachment of covers").



- (2) Pull the tip of the tool (a) to make sure that the checking roll slide smoothly without resistance.



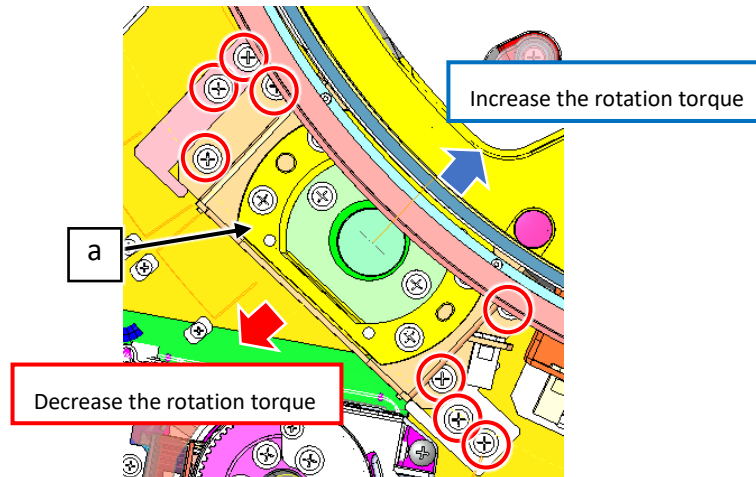
- (3) Remove the screw and loosen the screw, and then remove the cover (a) to access Motor unit (b).



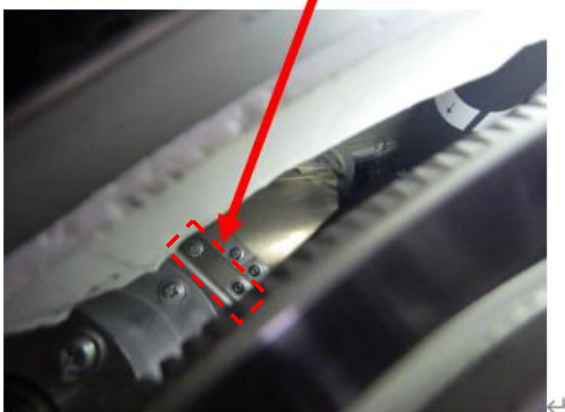
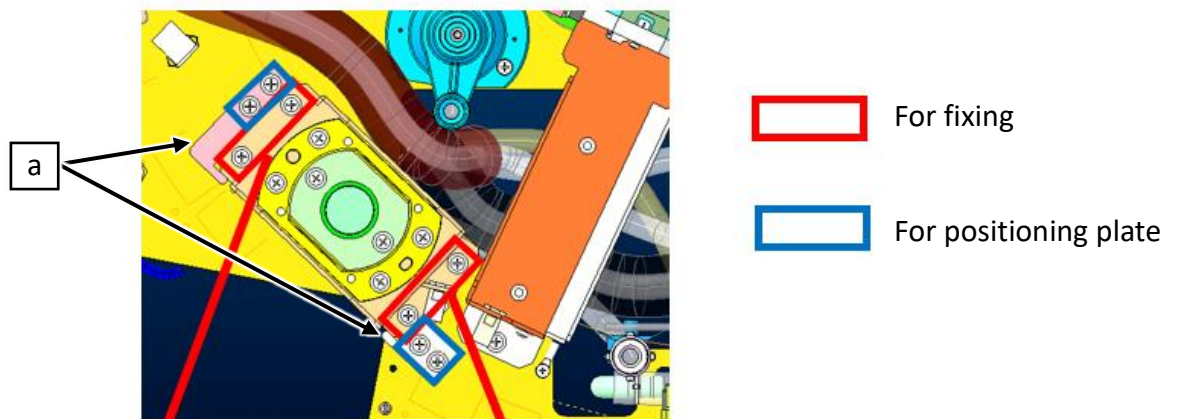
- (4) Loosen the eight screws and move Motor unit (a).

To increase the rotation torque, move Motor unit close to Reaction gear.

To decrease the rotation torque, move Motor unit away from Reaction gear.



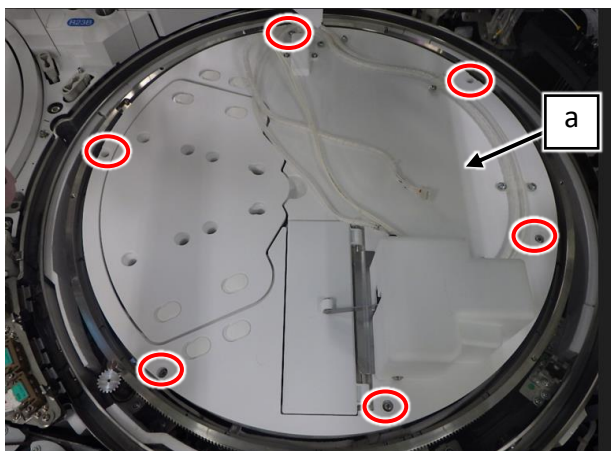
- (5) Align the 2 plates (a) in the blue area and tighten the 8 screws.



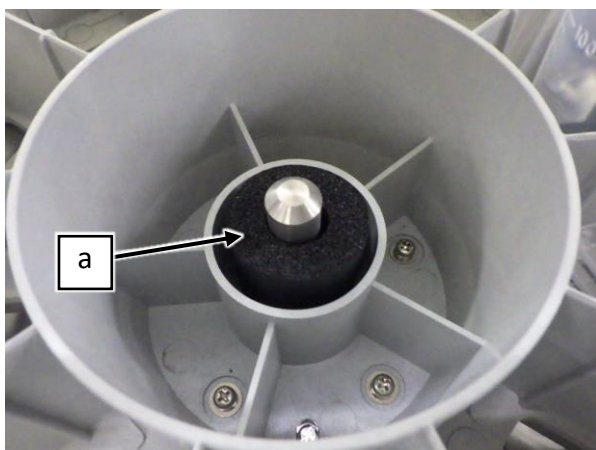
- (6) Retry from the section 3.1.

3.3 Adjustment of Reaction gear position

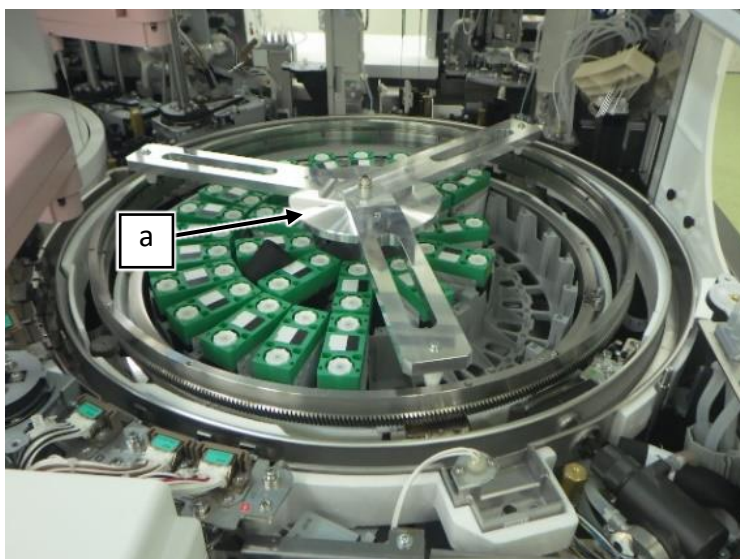
- (1) Loosen the 6 screws and remove Reagent jacket cover (a).



- (2) Remove the insulation tube (a).

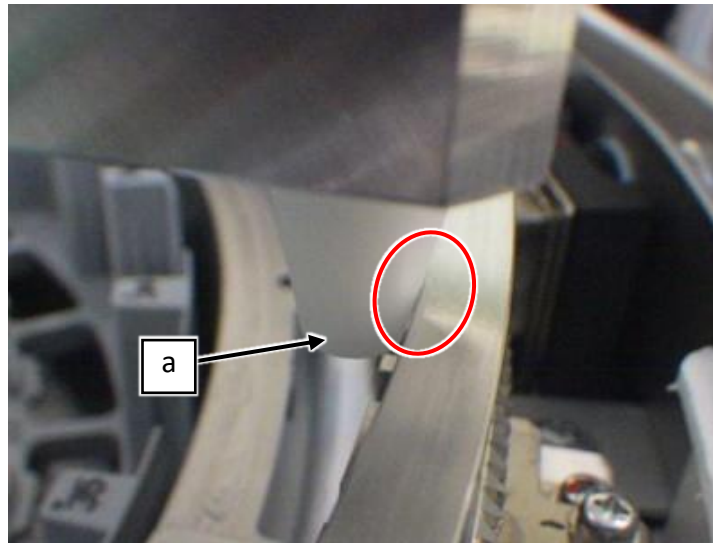


- (3) Set the tool (a).

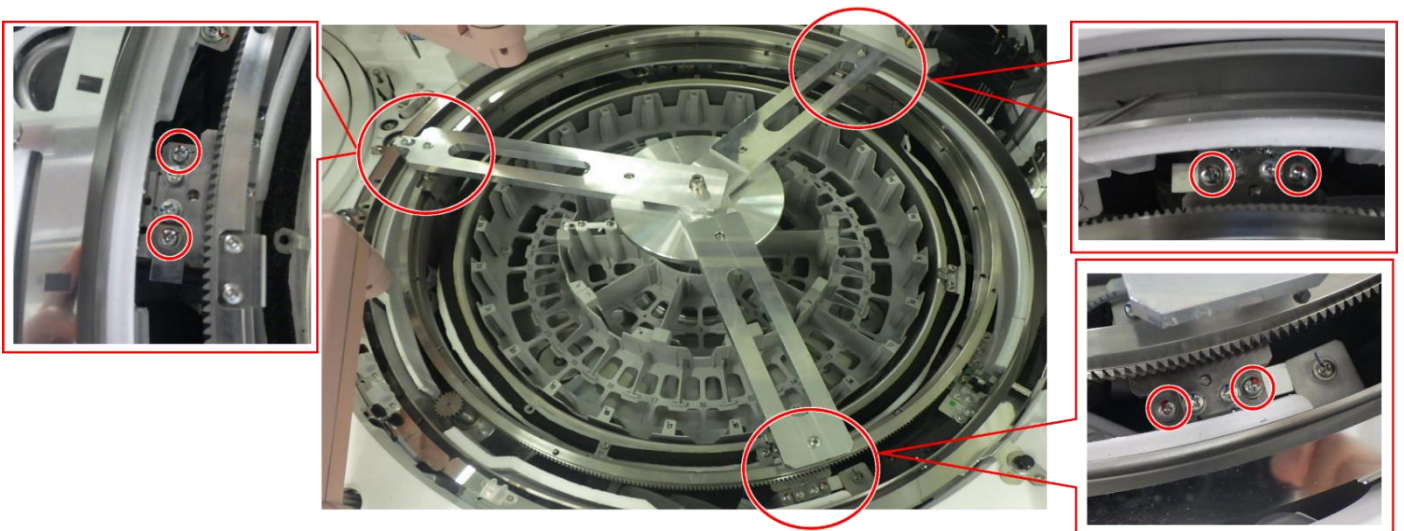


(a): TOOL 703 GEAR ADJUSTMENT (P/N: 848-3870)

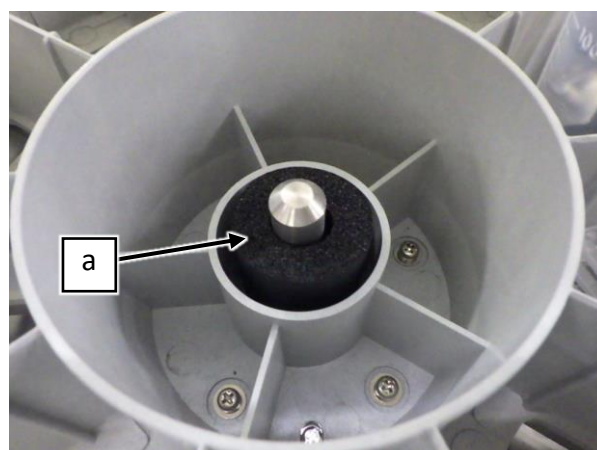
- (4) Make sure the 3 resin pins (a) contact Reaction gear.
If they contact, proceed to the step (7).
If not, proceed to the step (5).



- (5) Loosen the 6 screws and move Reaction gear for 3 resin pins (a) to contact Reaction gear.



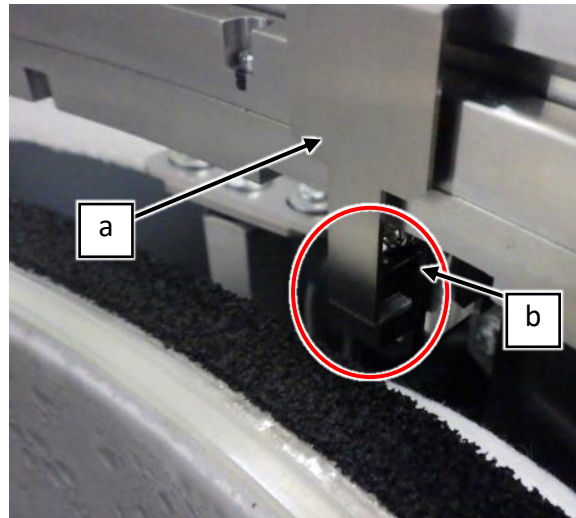
- (6) Tighten the 6 screws.
(7) Remove the tool and insert the insulation tube (a).



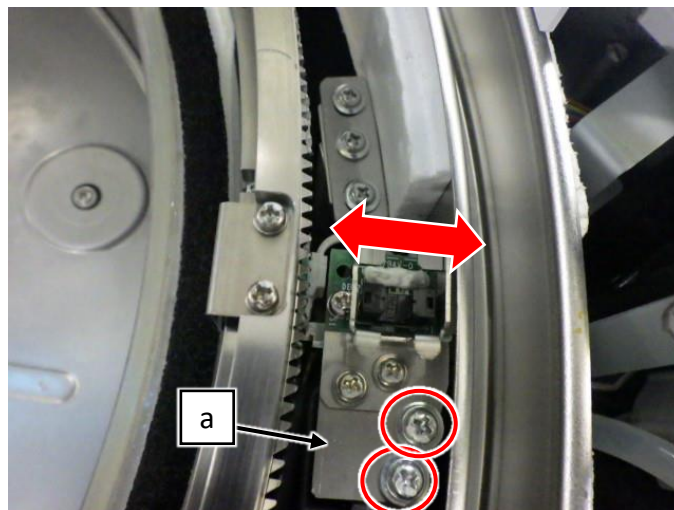
- (8) Rotate Reaction gear counterclockwise slowly and make sure that the detection plate (a) passes through the sensor (b) without contact.

If it passes through without contact, proceed to the step (9).

If not, proceed to the section 3.4.



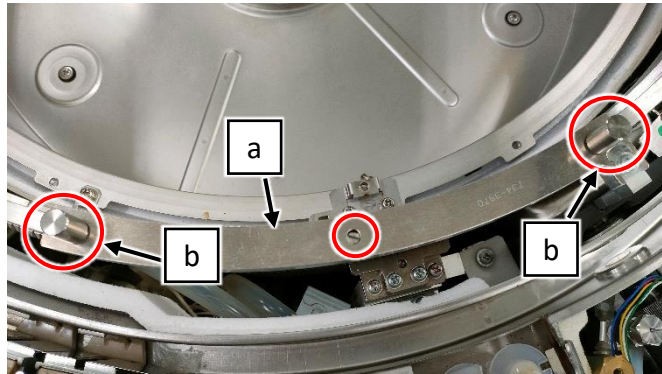
- (9) Loosen the 2 screws and slide the plate (a) not to contact the tool.



- (10) Tighten the 2 screws.

3.4 Adjustment(check) of count detection sensor position

- (1) Set the tool (a) to align the tool hole and the pin on Reaction gear set the 2 pins (b).



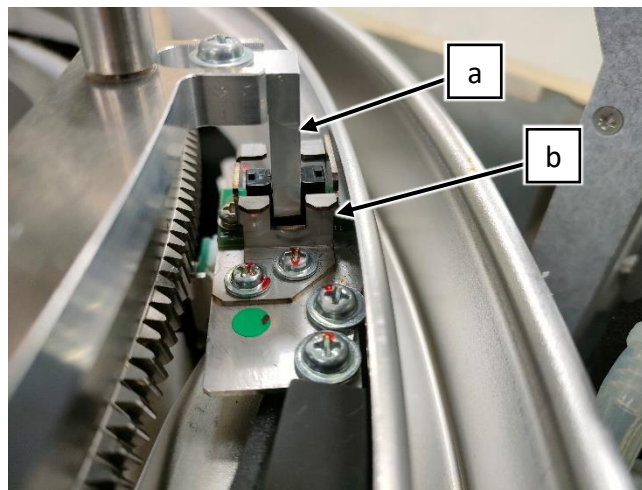
(a): TOOL DETECTOR ADJ (P/N: 734-3970)

(b): GEAR PIN (P/N: 734-1465)

- (2) Rotate Reaction gear counterclockwise slowly and make sure that the claw of the tool (a) passes through the sensor slit (b) without contact.

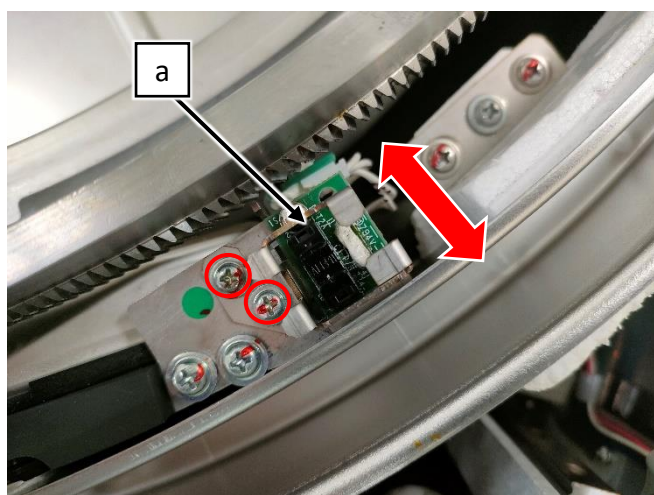
If it passes through without contact, proceed to the step (5).

If not, proceed to the step (3).

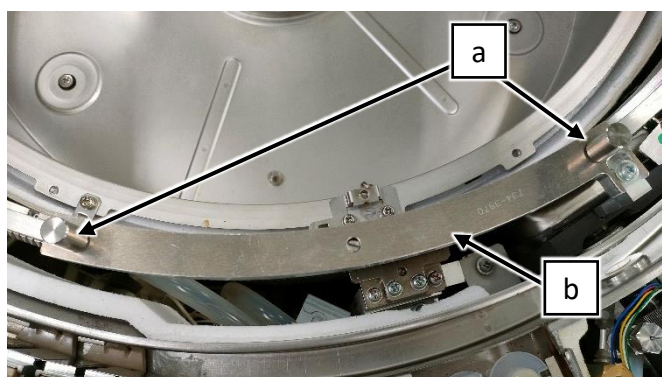


(a): TOOL DETECTOR ADJ (P/N: 734-3970)

- (3) Loosen the 2 screws and slide the sensor slit (a) not to contact the tool.



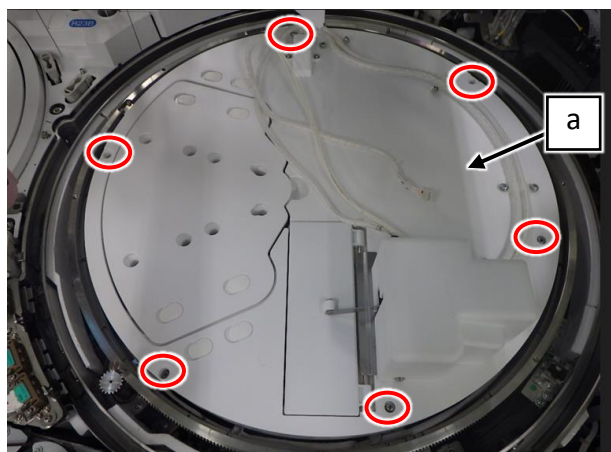
- (4) Tighten the 2 screws.
- (5) Remove the 2 pins (a) and the tool (b).



(a): GEAR PIN (P/N: 734-1465)

(b): TOOL DETECTOR ADJ (P/N: 734-3970)

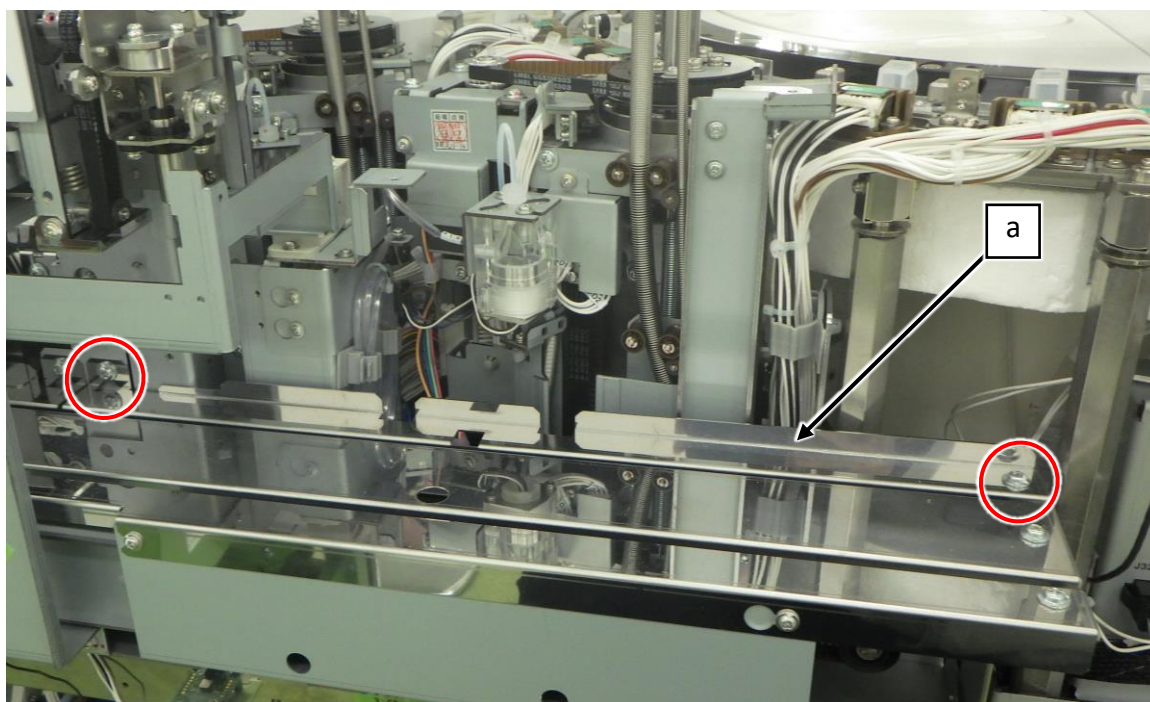
- (6) Attach Reagent jacket cover (a) and tighten the 6 screws.



- (7) Retry from the section 3.1.

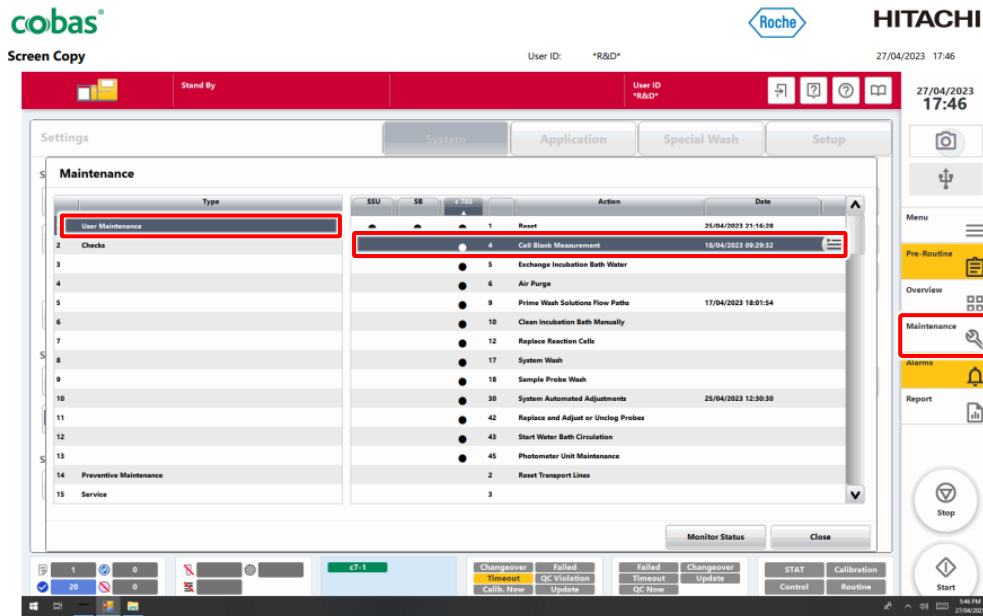
4. Accomplishment

- (1) Attach the cover (a) and tighten the 2 screws.



- (2) Install Reaction disk (refer to "S06-07 Replace procedure for Reaction disk").
- (3) Install Reagent probe R1B and R2/3B (refer to "Replacing the reagent probe" in User guide.).
- (4) Attach the covers (refer to "S06-41 Removal and attachment of covers").
- (5) Close COVER TOP F ASSY and COVER TOP R ASSY.
- (6) Perform the adjustment of Sample probe (Cell (Normal) position, Cell (Diluent) position, Rinse Check position) (refer to "S03-02 Sample Probe Adjustment").
- (7) Perform the adjustment of Reagent probe (Cell (R1) position, Cell (R2) position, Cell Height) (refer to "S03-03 Reagent Probe Adjustment").
- (8) Perform the adjustment of Cell Rinse unit (refer to "S03-04 Rinse Mechanism Position Adjustment").
- (9) Perform the adjustment of USM unit (refer to "S03-07 Ultra sonic mixer Adjustment").
- (10) Perform the adjustment of Photometer (refer to "S03-17 Photometer height Adjustment").
- (11) Perform ADC Photoelectric Timing Adjustment (refer to "S03-13 ADC Photoelectric Timing Adjustment").

(12) After ADC Photoelectric Timing Adjustment, Select [Maintenance] > [User Maintenance] > [Cell Blank Measurement].



(13) Select [c703] > [Perform].

(14) After Cell Blank Measurement, make sure no alarm occurs.

(15) Perform Instrument Check (refer to "S02-03 Instrument check").

End of Document.